

Final Year Project

Software Requirements Specifications

For

An Expert-Guided Multimodal AI Ecosystem for Diagnostic Intelligence

by

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1 Introduction

1.1 Purpose

This document details the functional and non-functional requirements for the **Expert-Guided Multimodal AI Ecosystem for Diagnostic Intelligence**, Version 1.0.

The purpose of this system is to provide an integrated, end-to-end platform for the analysis of brain tumors from multimodal Magnetic Resonance Imaging (MRI) scans. It addresses critical gaps in existing medical AI tools, such as performance degradation on new data and fragmented workflows, by unifying three core innovations:

- An advanced, multimodal tumor segmentation model (MoME+) enhanced with **continual learning** to ensure robust performance across diverse and evolving datasets.
- A structured reasoning module that translates segmentation outputs into quantitative, anatomically-grounded **JSON mappings**.
- An evidence-grounded **diagnostic report generation** service, powered by a fine-tuned Large Language Model (LLM), that produces clinically coherent narratives with clear traceability to visual evidence.

This document is intended to provide a comprehensive guide for the project's developers, advisors, and evaluators, establishing a clear scope and a shared understanding of the system's capabilities and constraints.

1.2 Document Conventions

This document follows a standardized format to ensure clarity and consistency for all readers. The following conventions are used throughout this SRS:

- **Requirement Identification:** All functional and non-functional requirements are uniquely identified with a prefixed tag and a sequential number. This allows for clear traceability and referencing. The prefixes used are:
 - **REQ-:** Functional Requirement
 - **PERF-:** Performance Requirement
 - **SAFE-:** Safety Requirement
 - **SEC-:** Security Requirement
 - **BR-:** Business Rule
- **Priority Levels:** All system features in Section 4 are assigned a priority level to indicate their importance to the project's success. The levels are defined as **High**, **Medium**, and **Low**.
- **Typographical Conventions:** Key technical terms, component names, and requirement tags are formatted in **bold text** to draw the reader's attention and improve readability.

1.3 Intended Audience and Reading Suggestions

This document is intended for a diverse audience involved in the development, evaluation, and use of the AI ecosystem. The content is organized to allow each reader to efficiently access the information most relevant to them.

Intended Audience:

- **Project Developers (Ahmed Sultan, Farhan Kashif):** This SRS serves as the primary technical guide for designing, implementing, and testing the system. It defines the complete scope of work.
- **Project Advisors & KG Committee (Dr. Muhammad Moazam Fraz, Dr. Naseer Bajwa, Dr. Rafi Ullah Chamlavi, Dr. Qaiser Riaz):** This document provides a detailed overview of the project's objectives, features, and technical specifications, facilitating evaluation and guidance.
- **End-Users (Clinicians, Radiologists):** This group can review the document to understand the system's capabilities, user interface, and expected outputs, ensuring it meets clinical workflow needs.

Reading Suggestions:

- This document is structured to be read either sequentially or by section, depending on the reader's role.
- For a general overview, all readers should begin with:
 - **Section 1: Introduction:** Outlines the purpose and scope of the project.
 - **Section 2: Overall Description:** Provides a high-level perspective of the product's functions, user classes, and operating environment.
- For specific interests, readers should proceed as follows:
 - **Project Advisors and Evaluators** should focus on **Section 2 (Overall Description)**, **Section 4 (System Features)**, and **Section 5 (Other Non-functional Requirements)** to assess the project's technical depth, innovation, and completeness.
 - **Developers** should read the document in its entirety, with particular attention to **Section 3 (External Interface Requirements)**, **Section 4 (System Features)**, and **Section 5 (Other Nonfunctional Requirements)**, as these sections contain the detailed specifications needed for implementation.
 - **Clinicians and End-Users** will find **Section 2.2 (Product Functions)**, **Section 3.1 (User Interfaces)**, and **Section 4 (System Features)** most relevant for understanding how the system will function from a user's perspective.

1.4 Product Scope

This document specifies the requirements for the **Expert-Guided Multimodal AI Ecosystem for Diagnostic Intelligence**, a unified platform designed to assist clinicians in the analysis of brain tumors from multimodal MRI scans. The system’s core purpose is to bridge the gap between the technical outputs of AI models and actionable, interpretable clinical information by integrating adaptive segmentation with evidence-grounded report generation.

Key Objectives and Benefits

The primary goals of this project are to:

- **Enhance Diagnostic Efficiency:** Improve the speed and consistency of brain tumor analysis, reducing turnaround time and inter-observer variability in radiology workflows.
- **Ensure Model Adaptability:** Overcome the limitations of static AI models by implementing a continual learning framework, allowing the system to adapt to new scanners and protocols without catastrophic forgetting.
- **Improve Interpretability and Trust:** Increase clinician trust by generating transparent diagnostic reports where every statement is explicitly linked to quantitative evidence derived from the imaging data.
- **Provide an End-to-End Solution:** Deliver a single, integrated workflow that takes a user from an MRI scan to a finalized, clinician-ready diagnostic report with visualizations, rather than providing fragmented, standalone tools.

Scope of Features

The following features are within the scope of this project:

- **Multimodal Segmentation Engine:**
 - Implementation of a MoME+ segmentation backbone.
 - Segmentation of three key tumor subregions: whole tumor (WT), tumor core (TC), and enhancing tumor (ET).
 - Integration of continual learning strategies (e.g., Elastic Weight Consolidation, replay memory) to maintain performance on evolving datasets.
- **Structured Data Generation:**
 - A module to post-process segmentation masks and map them to a standard brain atlas.
 - Automatic generation of structured JSON files containing quantitative metrics such as lesion volume, anatomical location, and centroids.
- **Automated Report Generation:**
 - A fine-tuned, lightweight Large Language Model (MedAlpaca-7B) to convert the structured JSON data into coherent, radiology-style narrative reports.

- A mechanism to ensure every sentence in the report is traceable to its source evidence in the JSON, enhancing explainability.
- **Clinician Web Portal:**
 - A secure web interface for clinicians to upload multimodal MRI scans.
 - Interactive visualization tools, including 2D overlays on MRI slices and 3D renderings of segmented tumors.
 - A user interface to review, edit, and finalize the AI-generated reports, with a feedback mechanism for future model refinement.
 - Functionality to export the final report, including text and visualizations, as a PDF document.
- **Deployment:**
 - The entire system will be containerized using Docker and deployed on a cloud platform (GCP) for accessibility.
 - Inference pipelines will be optimized for performance using quantization and specialized run-times.

Out of Scope

The following items are explicitly out of scope for Version 1.0 of this project:

- This system is intended as a diagnostic **support tool** and is not a substitute for a qualified clinician’s final diagnosis.
- The system will only process brain MRI scans and will not support other imaging modalities (e.g., CT, PET) or other parts of the body.
- Direct integration with hospital information systems (HIS) or Picture Archiving and Communication Systems (PACS) is not included.
- The project does not include seeking or obtaining regulatory or clinical approval (e.g., FDA, CE mark) for use in patient care.

1.5 References

1.5.1 Datasets

- [1] S. Bakas et al., “The multimodal brain tumor segmentation challenge 2024 (brats 2024),” in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Data available at <https://www.synapse.org/brats2024>, 2024.
- [2] S. Bakas et al., “The brats 2024 meningioma radiotherapy planning (brats-men-rt) challenge,” in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Data available at <https://www.synapse.org/brats2024>, 2024.

- [3] S. Bakas et al., “The brain tumor segmentation challenge 2023: Glioma segmentation,” in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Data available at <https://www.synapse.org/brats2023>, 2023.
- [4] S. Bakas et al., “The asnr-miccai brain tumor segmentation challenge 2023: Meningioma,” in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Data available at <https://www.synapse.org/brats2023>, 2023.
- [5] H. J. Kuijf et al., “Standardized assessment of automatic segmentation of white matter hyperintensities and results of the wmh segmentation challenge,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 11, pp. 2556–2568, 2019. DOI: 10.1109/TMI.2019.2912825.
- [6] S. Pati et al., *The university of california san francisco brain metastases stereotactic radiosurgery mri dataset*, <https://imagingdatasets.ucsf.edu/dataset/1>, Accessed 2025, 2023.
- [7] J. C. Peeken et al., *Brainmetshare: An international, multi-institutional mri dataset for brain metastases segmentation*, <https://aimi.stanford.edu/datasets/brainmetshare>, Accessed 2025, 2023.
- [8] S. Pati et al., “Federated tumor segmentation (fets) challenge 2022,” in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Data available at <https://www.synapse.org/fets>, 2022.

1.5.2 Papers

- [1] X. Zhang et al., “A foundation model for brain lesion segmentation with mixture of modality experts,” *arXiv preprint arXiv:2405.10246*, 2024, Accessed 2025.

2 Overall Description

2.1 Product Perspective

The Expert-Guided Multimodal AI Ecosystem for Diagnostic Intelligence is a new, self-contained software product designed to function as an advanced diagnostic support tool for clinicians analyzing brain tumors. It originates from the need to address critical limitations in current medical AI systems, which are often static, fragmented, and lack robust interoperability.

Unlike existing tools that treat segmentation and reporting as separate tasks, this product provides a single and unified pipeline that integrates these functions. It is not a replacement for an existing system but rather a novel solution that combines three key areas of research; multimodal segmentation, continual learning, and LLM-based report generation, into one cohesive workflow.

The system operates independently, but is designed for a clinical environment. It takes multimodal MRI scans as input and produces high-level diagnostic intelligence—including segmentation masks, quantitative data, and narrative reports as output for review by medical professionals.

A high-level view of the system’s architecture, illustrating the flow from the segmentation model to the final generated report, is provided in Figure 1.

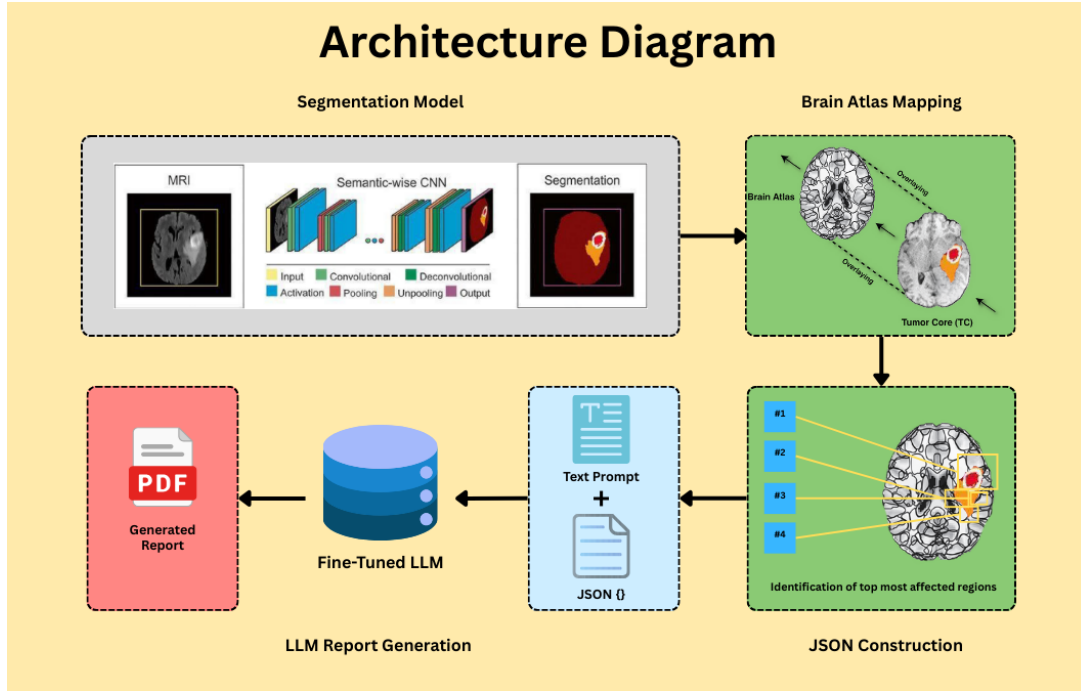


Figure 1: System Architecture Diagram, showing the main components and data flow from the Segmentation Model to the final Generated Report.

2.2 Product Functions

The system provides a comprehensive, end-to-end workflow for AI-assisted brain tumor analysis. The high-level flow of data through the system's core functions is illustrated in the Data Flow Diagram in Figure 2. The interactions that different users can have with these functions are summarized in the Use Case Diagram in Figure 3.

The primary functions are detailed below:

- **MRI Data Ingestion and Preprocessing:** The system shall allow users to upload multimodal brain MRI scans. It will then automatically perform necessary preprocessing steps, such as intensity normalization and bias-field correction, to prepare the data for analysis.
- **Automated Tumor Segmentation:** The system shall employ a deep learning model (MoME+) to automatically identify and delineate brain tumor subregions (Whole Tumor, Tumor Core, and Enhancing Tumor) from the preprocessed MRI data.
- **Continual Model Adaptation:** The underlying segmentation model is designed with a continual learning framework. This allows the system to be updated with new data from different scanners or protocols while mitigating the risk of catastrophic forgetting, thus ensuring its performance remains robust over time.
- **Quantitative Analysis and Structured Reporting:** After segmentation, the system shall analyze the tumor masks, map them to a standard brain atlas, and generate a structured JSON file containing key quantitative metrics like tumor volume, location, and centroids.

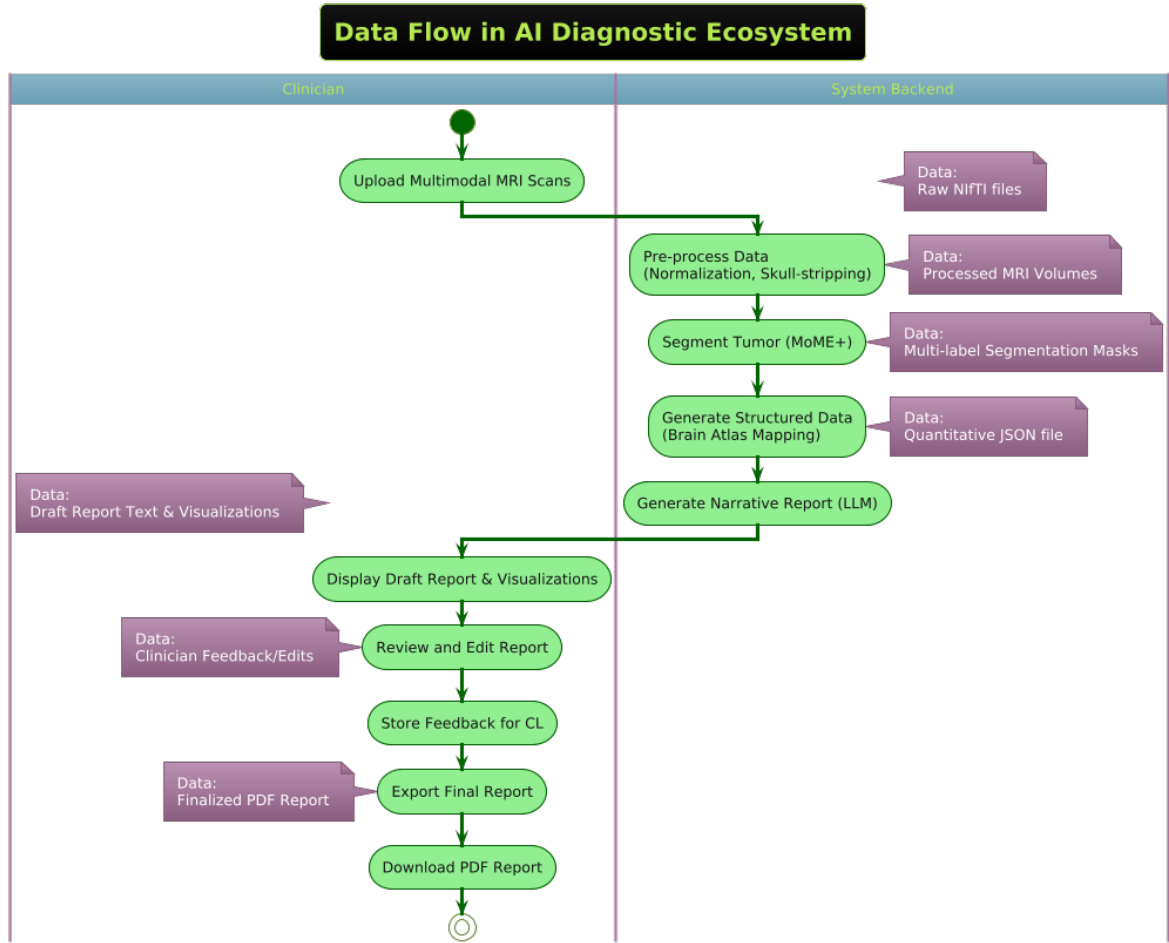


Figure 2: A high-level Data Flow Diagram (DFD) illustrating the journey of data from initial MRI upload to the final, exported PDF report.

- **Narrative Report Generation:** Using the structured JSON file as a factual basis, the system shall generate a coherent, clinically-styled narrative report that describes the findings in a human-readable format, ensuring all statements are grounded in the quantitative data.
- **Interactive Visualization and Review:** The system shall provide a web-based interface for clinicians to visualize the segmentation results, including 2D overlays on MRI slices and an interactive 3D rendering of the tumor. Clinicians will also be able to review and edit the generated narrative report.
- **Final Report Export:** Users shall be able to export the complete, finalized diagnostic report, which includes the narrative findings and key visualizations, into a standard PDF format.

2.3 User Classes and Characteristics

The system is designed to serve several distinct classes of users, each with different needs and levels of technical expertise. The primary user class is the clinical professional, for whom the system is designed as a daily-use diagnostic aid.

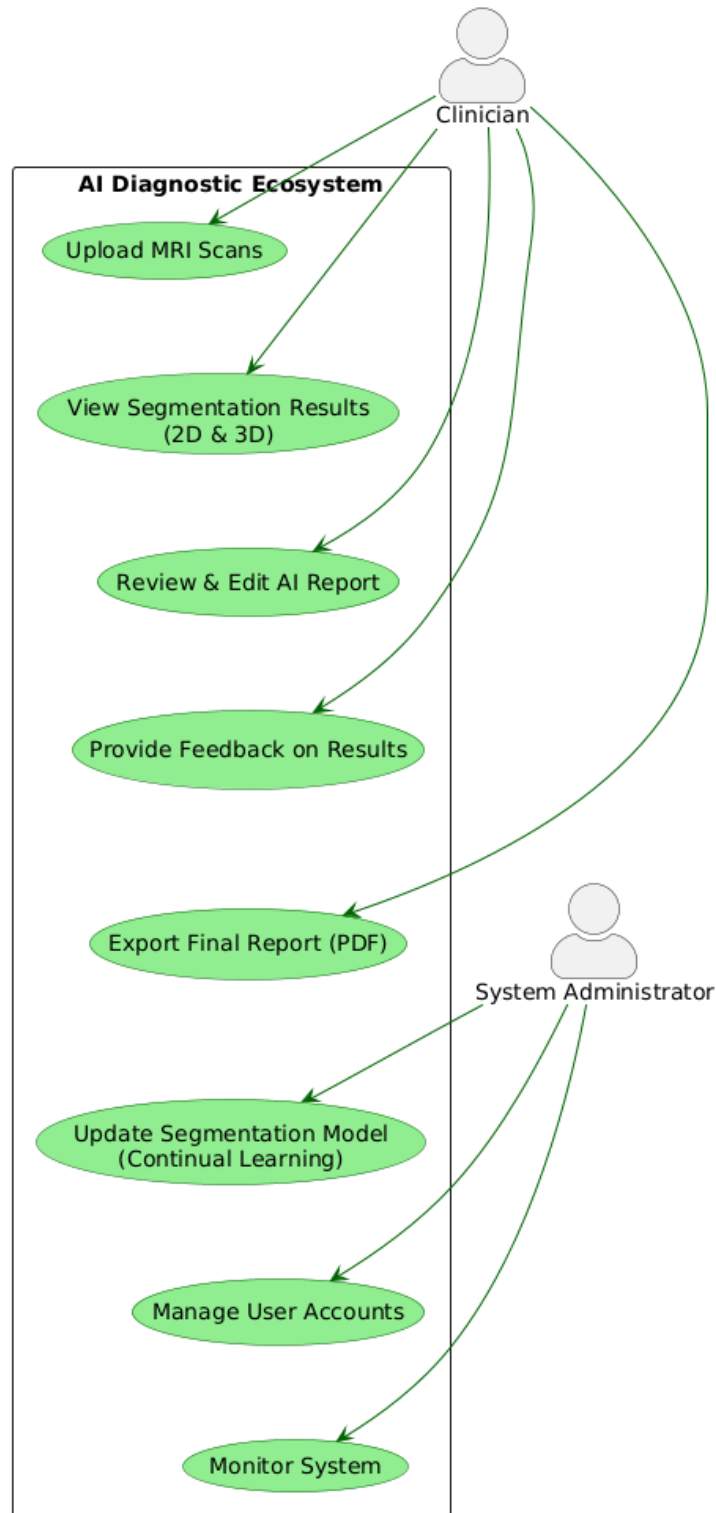


Figure 3: A Use Case Diagram showing the primary functions available to the Clinician and System Administrator roles.

Primary User Class: Clinicians

This group includes radiologists, neuro-oncologists, and surgeons who are directly involved in patient diagnosis and treatment planning. They are the most important user class for this product.

- **Characteristics:** These users possess high-level medical expertise but may have varying levels of technical proficiency with AI software. They require tools that are efficient, reliable, and easily integrated into their diagnostic workflow.
- **Needs:** Their primary need is for accurate, timely, and consistent tumor segmentation to guide surgical planning, radiotherapy, and monitoring. They require interpretable and trustworthy outputs, including clear visualizations and auditable reports where findings are explicitly linked to evidence.

Secondary User Class: Medical Researchers

This class consists of users within clinical research consortia and multi-site trials who require standardized data analysis tools.

- **Characteristics:** These users are technically proficient and focus on quantitative data analysis and biomarker extraction.
- **Needs:** They require a robust and adaptive pipeline that can harmonize data processing across different acquisition sites, scanners, and protocols. The continual learning feature is particularly important for maintaining model consistency as new data is added over the course of a study.

System Maintenance Class: Developers and Administrators

This group includes the project team (Ahmed Sultan, Farhan Kashif) and any future personnel responsible for maintaining and updating the system.

- **Characteristics:** These users have deep technical expertise in machine learning, software development, and system deployment (Python, PyTorch, Docker, GCP).
- **Needs:** They require access to the backend for model updates, performance monitoring, and managing the continual learning loop (i.e., retraining the model with feedback data). They are also responsible for managing the system’s deployment and documentation.

Indirect User Class: Patients

While not direct operators of the system, patients are the ultimate beneficiaries of the technology.

- **Characteristics:** This group has no medical or technical expertise.
- **Needs:** They require clear, understandable, and accurate diagnostic information. The system addresses this by providing concise, patient-friendly PDF reports that combine text with 2D and 3D visualizations of the findings.

2.4 Operating Environment

The Expert-Guided Multimodal AI Ecosystem for Diagnostic Intelligence is designed as a cloud-deployed web application. The system will operate in a client-server environment, requiring a modern web browser on the client side and a specific stack of technologies on the server side.

Server-Side Environment

The backend and machine learning models will be deployed within Docker containers on Google Cloud Platform (GCP), providing a consistent and scalable environment.

- **Operating System:** The Docker containers will be based on a standard Linux distribution (e.g., Ubuntu 22.04 LTS).
- **Backend Framework:** The REST API will be served using either FastAPI or Django REST Framework.
- **Asynchronous Task Queue:** Celery with a Redis message broker will be used to manage long-running jobs such as model inference and report generation.
- **ML/AI Frameworks:**
 - PyTorch, supplemented by the MONAI library for segmentation tasks and continual learning implementation.
 - The Hugging Face Transformers library for fine-tuning and serving the MedAlpaca-7B language model, using the PEFT (LoRA) framework for efficiency.
- **Inference Optimization:** Inference will be optimized using runtimes like ONNX Runtime or vLLM and quantization techniques such as bitsandbytes or GPTQ to ensure acceptable performance.
- **Database:** A PostgreSQL database will be used for storing user data, metadata for MRI scans, and other relational information. Redis will be used for caching.

Client-Side Environment

The user-facing application is a web portal accessible through a web browser.

- **Supported Browsers:** The application will be designed to be compatible with the latest versions of modern web browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge.
- **Frameworks:** The frontend will be built using React.js with Tailwind CSS for styling and Three.js for interactive 3D tumor visualization.

Hardware Environment

- **Training:** Model training and fine-tuning will be performed on relatively high-performance machines equipped with NVIDIA GPUs (e.g. gtx 4060, a10) to handle the computational load.
- **Deployment (Server):** The deployed system will run on Google Cloud Platform (GCP) virtual machine instances equipped with GPUs suitable for inference (e.g. NVIDIA T4).
- **Deployment (Client):** No specific hardware is required on the client side beyond a standard computer capable of running a modern Web browser.

2.5 Design and Implementation Constraints

The development and implementation of this system are subject to several constraints that will guide the design choices and technical approach.

- **Technology Stack:** The project must adhere to the specified technology stack. The machine learning components must be developed in Python using PyTorch and the Hugging Face Transformers library. The backend API must be built with either FastAPI or Django, and the front-end must use React.js. This ensures the project team can leverage their existing expertise.
- **Open-Source Preference:** To manage costs and ensure transparency, the project will prioritize the use of open-source software and libraries (e.g., MONAI, PyTorch, Hugging Face, Celery, Redis, PostgreSQL) wherever feasible.
- **Hardware and Cost Limitations:** The project operates under a defined budget of PKR 60,000. This constrains the resources available for cloud computing (GCP storage and GPU instances) and dictates the need for efficient model design. Consequently, the project will use lightweight, parameter-efficient models (e.g., MedAlpaca-7B with LoRA) and optimization techniques like quantization to minimize deployment costs.
- **Data Availability:** The system’s performance is fundamentally dependent on the availability and quality of public, curated multimodal MRI datasets. The project will leverage data from several established sources, including the **BraTS Challenges (2023, 2024)**, the **FeTS Challenge (2022)**, the **WMH Segmentation Challenge**, and datasets focused on brain metastases such as **UCSF-BMSR** and **BrainMetShare**. Consequently, a significant design constraint is the need to build a system robust enough to handle the considerable heterogeneity in imaging protocols, scanner types, and lesion characteristics present across these diverse sources.
- **Platform Independence (Client):** The user-facing portal must be a web-based application accessible via standard modern browsers, avoiding the need for native client-side software installations.
- **No PACS/HIS Integration:** Version 1.0 of the system will not integrate directly with hospital Picture Archiving and Communication Systems (PACS) or Hospital Information Systems (HIS). Data upload will be a manual process, which constrains the system’s workflow to a standalone model.
- **Non-Clinical Grade:** The system is being developed as a final year project and a proof-of-concept. It is not intended for real-time clinical diagnosis and will not undergo regulatory approval processes (e.g., FDA). This constrains the system to be used for research and support purposes only.

2.6 User Documentation

To ensure the effective use and future maintenance of the system, a set of documentation will be provided along with the final product. All documentation will be written collaboratively using Overleaf and delivered in PDF format, as outlined in the project management plan.

The following user documentation components will be delivered:

- **User Manual for Clinicians:** A comprehensive guide intended for the primary end users (clinicians, radiologists). This document will provide step-by-step instructions on:
 - Accessing the web portal and management user accounts.
 - The required format and procedure for uploading multimodal MRI scans.
 - Navigating the user interface to view and interact with the 2D and 3D segmentation visualizations.
 - Reviewing, editing, and interpreting AI-generated diagnostic reports.
 - Exporting the final report as a PDF document.
- **System Administrator and Deployment Guide:** A technical document for developers and system administrators responsible for deploying and maintaining the application. This guide covers:
 - System requirements and prerequisites for server-side deployment.
 - Step-by-step instructions for deploying the system using the provided Docker containers.
 - Configuration of the application, including database connections and model paths.
 - Instructions on how to manage the continual learning pipeline, including how to incorporate new data for model fine-tuning.
- **Quick Start Guide:** A concise, one-to-two-page document summarizing the core workflow for clinicians, enabling new users to quickly begin using the system for basic tasks.

2.7 Assumptions and Dependencies

This section lists the key assumptions made during the planning of this project and the external factors upon which the project depends. The project’s success could be affected if these assumptions prove to be incorrect or if the dependencies are not met.

Assumptions

- **Data Quality Assumption:** It is assumed that the public datasets identified for this project (e.g., BraTS, WMH) have sufficiently accurate annotations and image quality to train a robust and clinically relevant segmentation model.
- **Open-Source Software Stability:** The project assumes that the selected open-source libraries and frameworks (e.g., PyTorch, MONAI, Hugging Face Transformers, React.js) will remain stable, well-documented, and perform as expected throughout the development lifecycle.
- **Cloud Service Reliability:** It is assumed that the Google Cloud Platform (GCP) services required for storage and computation will be consistently available and reliable during the project timeline.

- **Model Suitability:** The project assumes that the chosen foundational models, specifically MoME+ for segmentation and MedAlpaca-7B for report generation, are appropriate architectural choices that can be successfully fine-tuned to meet the performance requirements of the system.

Dependencies

- **Dependency on Public Datasets:** The project is fundamentally dependent on the continued availability and accessibility of the public multimodal MRI datasets required for training, validation, and testing of segmentation models.
- **Dependency on Third-Party Libraries:** The implementation is critically dependent on the functionality and APIs of several third-party open-source libraries, including PyTorch, MONAI, Hugging Face Transformers, FastAPI/Django, and React.js.
- **Dependency on Pre-trained Models:** The report generation component depends on the availability of the pre-trained model weights for MedAlpaca-7B, which serves as the base for the fine-tuning process.
- **Dependency on Cloud Infrastructure:** The project's deployment and parts of the development (heavy training) are dependent on the infrastructure provided by Google Cloud Platform, specifically GCP Buckets for storage and GPU-enabled virtual machine instances for computation.

3 External Interface Requirements

3.1 User Interfaces

The system will feature a single, web-based Graphical User Interface (GUI) designed for clinical professionals. The interface will be developed using React.js with Tailwind CSS for a clean, modern, and intuitive user experience. The primary goal of the UI is to provide a seamless workflow from scan upload to final report generation.

General GUI Standards

- **Layout and Design:** The interface will adopt a professional, uncluttered layout. It will be designed to be responsive and function correctly on standard desktop and laptop screen resolutions.
- **Consistency:** Navigation, button styles, terminology, and overall user interactions will be consistent across all views of the application.
- **Feedback:** The system will provide clear visual feedback, including loading indicators for long-running processes and informative messages for successful operations or errors.

Key Screens and Wireframes

The user's workflow is organized into four key screens, presented in the wireframes below.

1. **Login Page:** The user's entry point to the system, ensuring secure access via authentication. (See Figure 4).

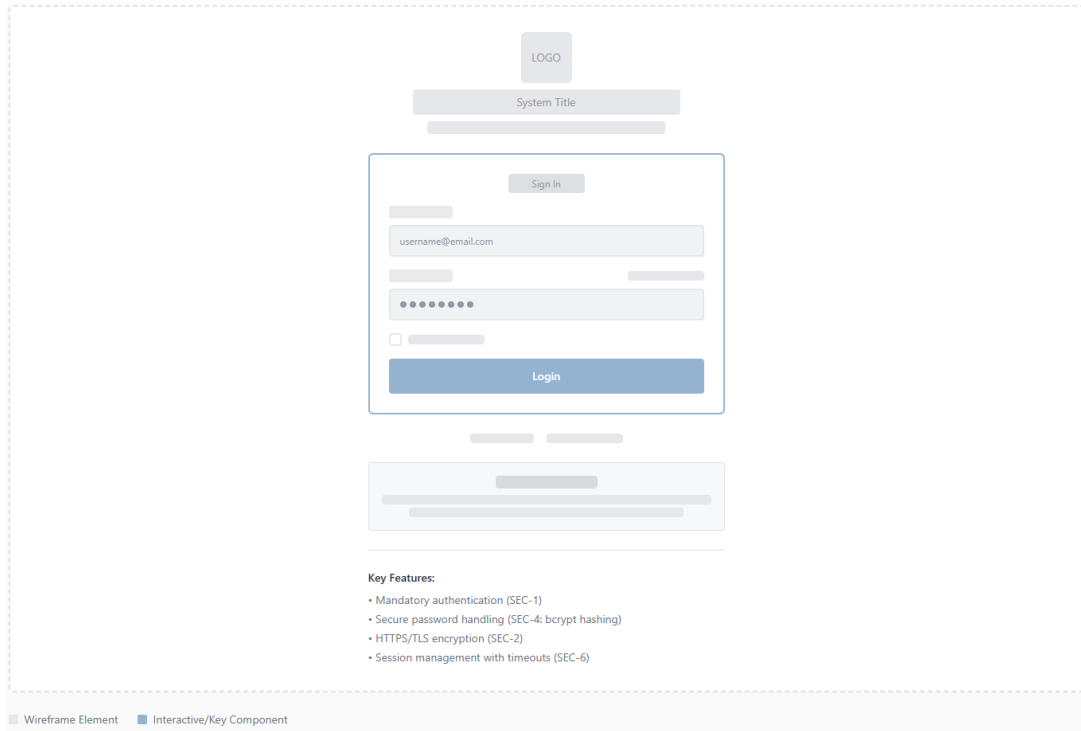


Figure 4: Wireframe of the Login Page.

2. **Scan Upload Page:** After logging in, the user is presented with the main dashboard. This view allows the user to select and upload the required multimodal MRI scans and view the status of previously uploaded scans. (See Figures 5 and 6).

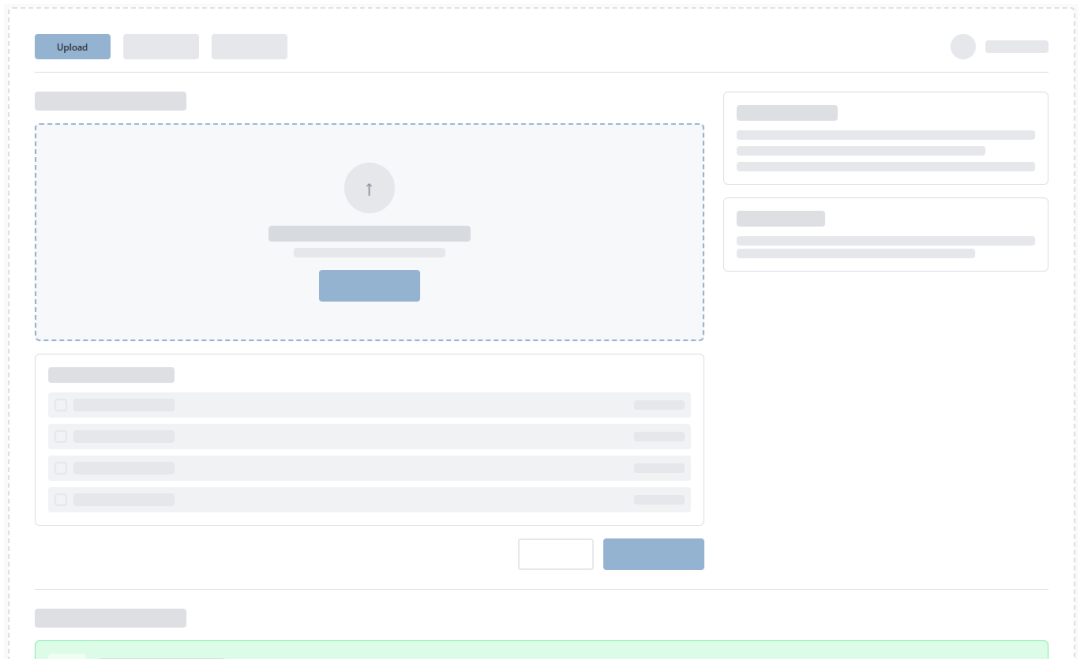


Figure 5: Wireframe of the Scan Upload Page (Part 1 of 2).

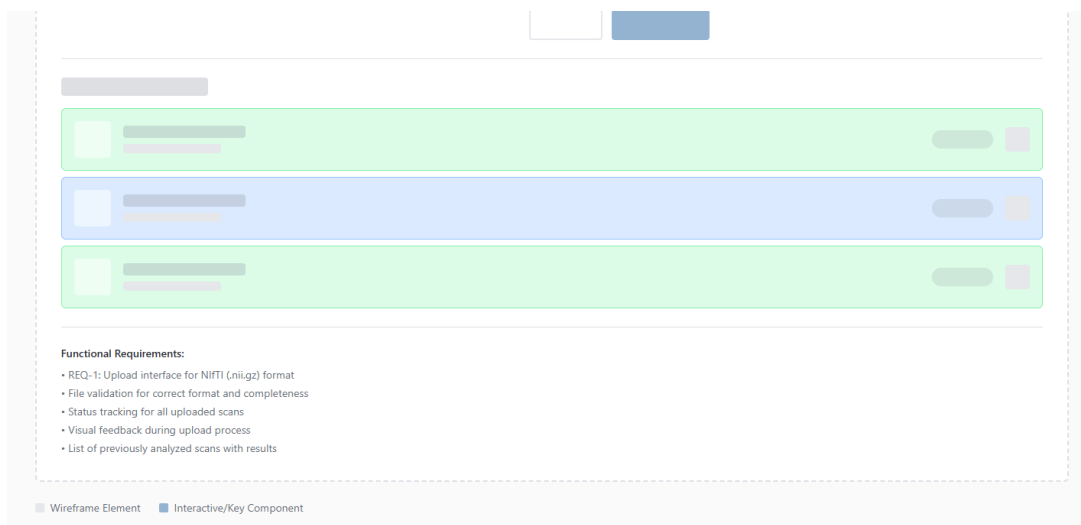


Figure 6: Wireframe of the Scan Upload Page (Part 2 of 2).

3. **Analysis Dashboard:** This is the main interactive view where the AI-generated results are presented. It features panels for 2D slice viewing, 3D tumor model interaction, and the editable narrative report. (See Figures 7 and 8).

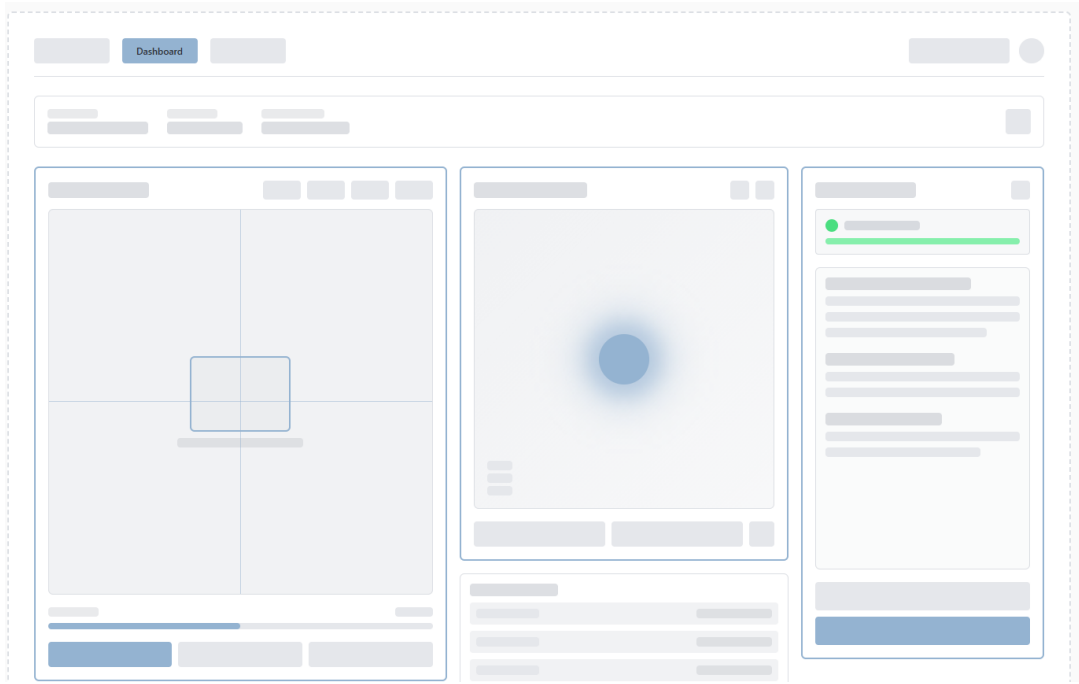


Figure 7: Wireframe of the Analysis Dashboard (Part 1 of 2).

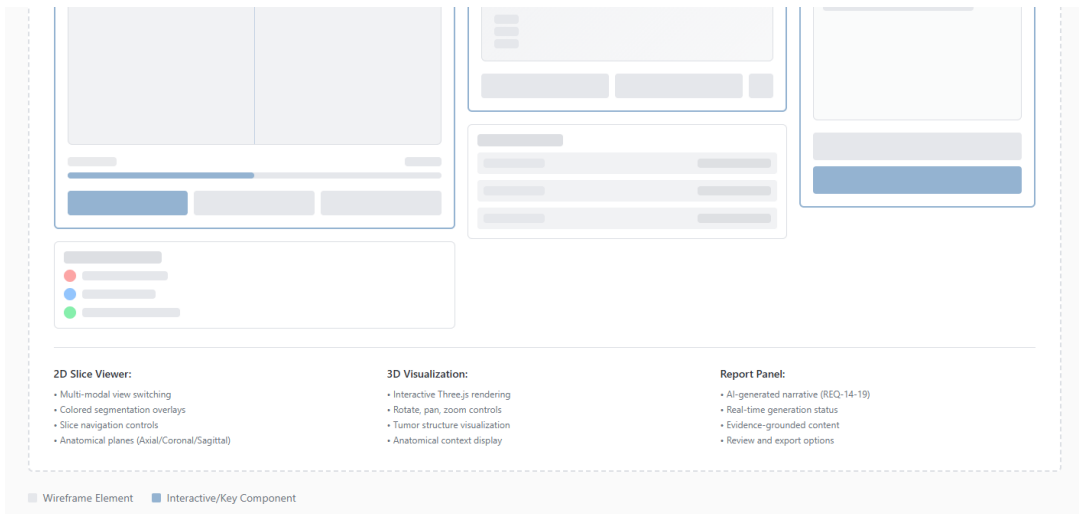


Figure 8: Wireframe of the Analysis Dashboard (Part 2 of 2).

4. **Final Report View:** This view shows the finalized, exportable PDF report, which combines the clinician-approved text with the key visualizations. (See Figures 9 and 10).

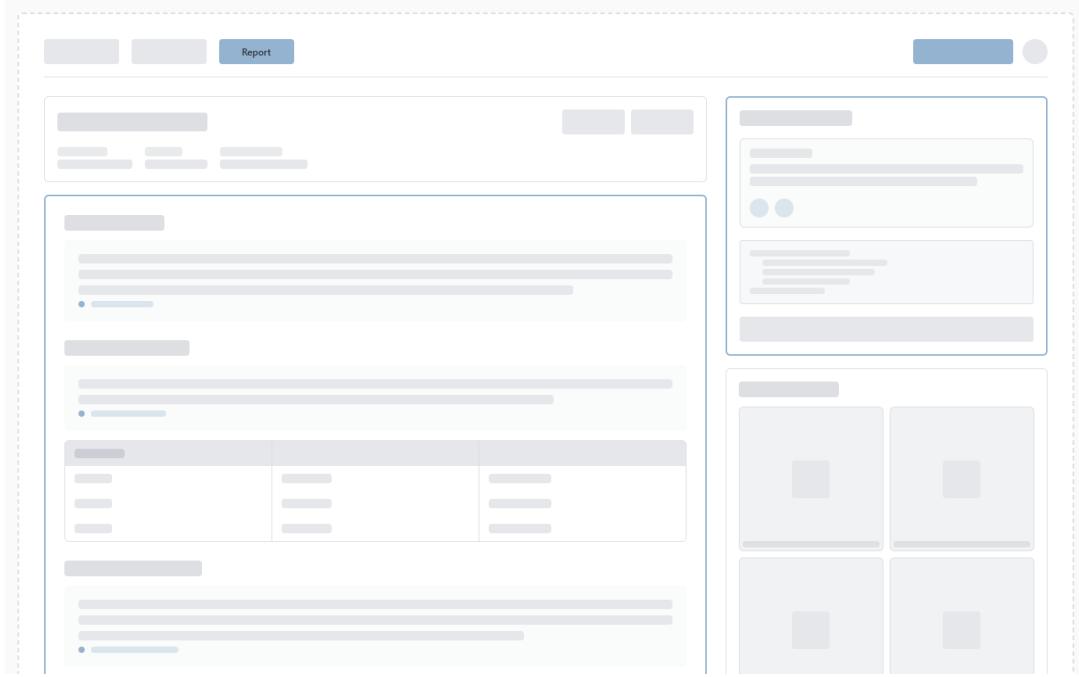


Figure 9: Wireframe of the Final Report View (Part 1 of 2).

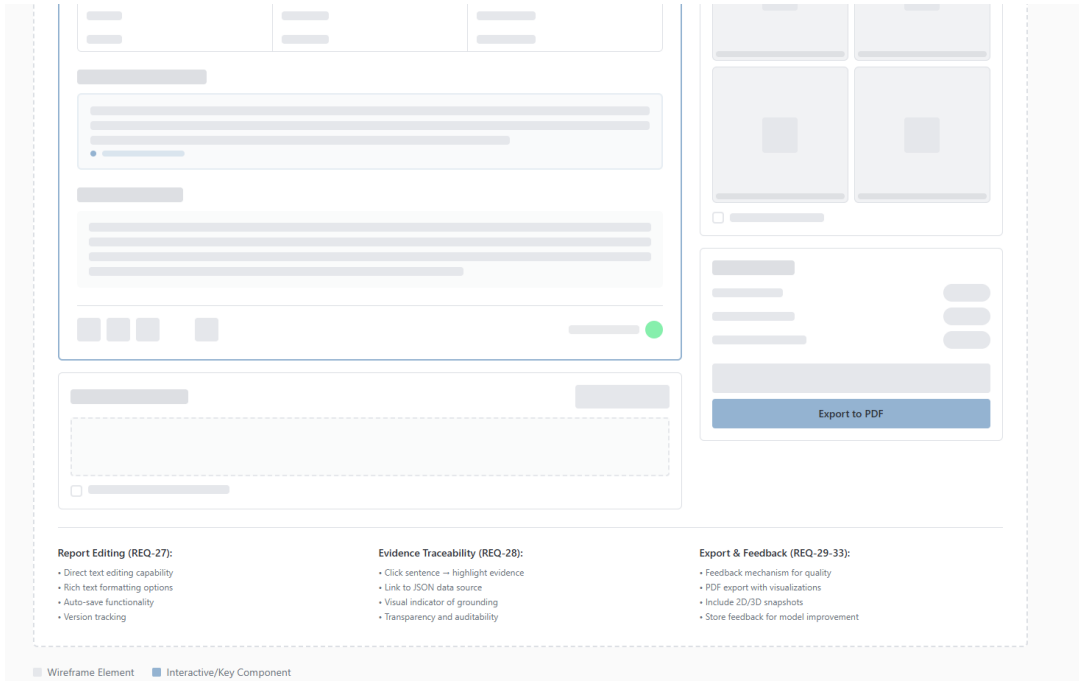


Figure 10: Wireframe of the Final Report View (Part 2 of 2).

3.2 Hardware Interfaces

The Expert-Guided Multimodal AI Ecosystem is a software-only system and does not have any direct physical or logical interfaces with custom or specialized medical hardware, such as MRI scanners. All data is expected to be provided to the system as digital files. The system’s interaction with hardware is limited to standard computing components, managed through the operating system and established software drivers.

- **Server-Side Hardware:** The backend and machine learning components require high-performance Graphics Processing Units (GPUs) for training and inference. The software interacts with these GPUs through standard, high-level libraries and APIs (e.g., NVIDIA CUDA), not through a custom-defined hardware interface.
- **Client-Side Hardware:** The user-facing web application runs on standard desktop or laptop computers. All interactions with client-side hardware, such as displays, keyboards, and mice, are managed by the end-user’s web browser and operating system. No specific hardware interface requirements are defined for the client.

3.3 Software Interfaces

The system is built upon a modern technology stack, interfacing with a variety of open-source software components, libraries, and tools. These interfaces are managed through well-defined APIs and SDKs.

Application Programming Interface (API)

The primary software interface for the system is a RESTful API developed using either FastAPI or the Django REST Framework.

- **Description:** This API exposes the core functionalities of the system (scan upload, segmentation, report generation) to the frontend client. The frontend communicates with the backend exclusively through this API, sending HTTP requests and receiving JSON responses.
- **Data Items:**
 - **Incoming:** Multimodal MRI scans (in NIfTI format) are sent from the client to the server.
 - **Outgoing:** The server sends status updates, structured JSON data containing analysis results, and pointers to visualization assets back to the client.

Machine Learning and AI Libraries

The core AI functionalities are implemented by interfacing with several specialized Python libraries:

- **PyTorch & MONAI:** The segmentation and continual learning models are built using PyTorch. The MONAI library provides domain-specific tools and pre-built components for medical imaging tasks, which are used to accelerate development.
- **Hugging Face Transformers & PEFT:** The system interfaces with the Transformers library to load, fine-tune, and run inference with the MedAlpaca-7B language model. The Parameter-Efficient Fine-Tuning (PEFT) library is used to manage the LoRA fine-tuning process.
- **ONNX Runtime / vLLM:** To optimize inference speed, the system will interface with a high-performance serving runtime. The PyTorch models will be exported to the ONNX format or served via vLLM.

Database and Storage Interfaces

- **PostgreSQL Database:** The application interfaces with a PostgreSQL database via a standard Python database adapter (e.g., `psycopg2`). It is used to store and retrieve all relational data, such as user information and metadata for each analysis case.
- **Redis:** The system connects to a Redis server for two purposes: as a message broker for the Celery asynchronous task queue and as a high-speed cache for session data or frequently accessed information.
- **Cloud Storage (GCP Buckets):** The application interfaces with Google Cloud Storage via the GCP SDK. This interface is used for the persistent storage of large binary files, including the uploaded MRI volumes and the generated segmentation masks.

Operating System

The application is designed to be deployed within Docker containers running a Linux-based operating system (e.g., Ubuntu). The software interfaces with the OS for fundamental operations like file system access, process management, and networking, but it does not depend on any OS-specific features, ensuring portability.

3.4 Communications Interfaces

The system’s communication is primarily based on a standard client-server architecture over the internet. The following specifies the protocols and standards that govern data exchange for the application.

- **Network Communication Protocol:** All communication between the client-side user interface (web browser) and the backend server will use the Hypertext Transfer Protocol Secure (HTTPS). The use of HTTPS is mandatory to ensure that all data in transit, including sensitive medical images and user credentials, is encrypted using Transport Layer Security (TLS).
- **Message Formatting:** Data exchanged between the client and the server via the REST API will be formatted as JavaScript Object Notation (JSON). This lightweight and human-readable format will be used for API requests, responses, and for the structured data that grounds the LLM report generation.
- **Web Browser Interface:** The system interfaces with the user through a web browser. The application relies on the browser’s standard implementation of HTTPS and its ability to process HTTP requests and handle JSON data.
- **Communication Security:** Beyond the encryption provided by HTTPS, access to the system’s API endpoints will be protected by an authentication mechanism (e.g., token-based authentication) to ensure that only authorized users can upload data and access results.
- **Data Transfer and Synchronization:**

- Users require a stable, high-bandwidth internet connection to facilitate the upload of large, multimodal MRI files in a reasonable amount of time.
- The system employs a synchronous request-response model for immediate UI interactions. For time-consuming computational tasks such as segmentation and report generation, the system uses an asynchronous communication mechanism managed by a Celery and Redis task queue. This prevents the user interface from becoming unresponsive while waiting for analysis to complete.

4 System Features

This section details the functional requirements of the system, organized by major features. Each feature is described with its priority, stimulus-response sequences, and a list of specific functional requirements.

4.1 System Feature 1: Automated Multimodal Tumor Segmentation

4.1.1 Description and Priority

This feature describes the core functionality of the system: the automated analysis of user-uploaded multimodal MRI scans to identify and delineate brain tumor subregions. The system employs an advanced deep learning model (MoME+) to produce accurate segmentation masks that serve as the foundation for all subsequent analysis and report generation.

Priority: High. This is the primary and most critical feature of the entire ecosystem.

4.1.2 Stimulus/Response Sequences

1. **Stimulus:** The user selects the required multimodal MRI scan files (e.g., T1, T1ce, T2, FLAIR) via the upload interface and initiates the analysis process.

2. **System Response:**

- The system validates the uploaded files for correct format and completeness.
- It starts the segmentation process as an asynchronous background job.
- The user interface displays a status indicator to inform the user that analysis is in progress.
- Upon successful completion, the system presents the results on the visualization dashboard, displaying the 2D overlays and 3D tumor model.
- If the analysis fails, the system displays a clear error message to the user.

4.1.3 Functional Requirements

REQ-1: The system shall provide an interface for the user to upload multimodal brain MRI scans in the NIFTI (.nii.gz) file format.

REQ-2: The system shall automatically perform a series of preprocessing steps on the uploaded data, including skull-stripping, intensity normalization, and bias-field correction, prior to model inference.

REQ-3: The system shall utilize a Mixture-of-Modality-Experts (MoME+) architecture as the core segmentation model.

REQ-4: The segmentation model shall output multi-label masks identifying three distinct tumor subregions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

REQ-5: The segmentation process must be handled asynchronously to ensure the user interface remains responsive while the model is running.

REQ-6: The system shall display the generated segmentation masks as colored overlays on the corresponding 2D MRI slices in an interactive viewer.

REQ-7: The system shall generate and display an interactive 3D visualization of the segmented tumor subregions.

4.2 System Feature 2: Structured Data Generation and Brain Atlas Mapping

4.2.1 Description and Priority

This feature describes the crucial intermediate step of translating the raw segmentation masks into a structured, quantitative format. The system will post-process the masks, compute key metrics, and align the findings with a standard brain atlas to provide anatomical context. This structured data, formatted as a JSON object, serves as the audit-able and factual foundation for the narrative report generation feature.

Priority: High. This feature is the critical link between the computer vision output and the language model input, ensuring the final report is grounded in evidence.

4.2.2 Stimulus/Response Sequences

1. **Stimulus:** The automated segmentation process (Feature 4.1) completes successfully, producing multi-label tumor masks. This is an internal, server-side trigger.
2. **System Response:**
 - The system automatically initiates the post-processing and analysis of the segmentation masks.
 - It computes various quantitative metrics from the masks.
 - It aligns the masks with a pre-loaded digital brain atlas.
 - The system generates a structured JSON file containing all the derived information.
 - This JSON file is stored and becomes the input for the report generation module.

4.2.3 Functional Requirements

REQ-8: The system shall post-process the raw segmentation masks to calculate the volume (in cm^3) for each identified tumor subregion (WT, TC, ET).

REQ-9: The system shall compute the 3D centroid coordinates for each tumor subregion.

REQ-10: The system shall register the patient’s MRI and the corresponding segmentation masks to a standard brain atlas (e.g., MNI atlas).

REQ-11: The system shall determine and list the specific anatomical brain regions (e.g., left frontal lobe, right temporal lobe) that have been affected by the tumor based on the atlas registration.

REQ-12: All computed metrics and anatomical location data must be compiled and stored in a single, structured JSON file.

REQ-13: The schema for the output JSON shall be well-defined, versioned, and documented to ensure consistency for the downstream report generation module.

4.3 System Feature 3: Evidence-Grounded Diagnostic Report Generation

4.3.1 Description and Priority

This feature describes the system’s ability to automatically generate a human-readable, clinically relevant diagnostic report. Using the structured JSON data generated in Feature 4.2 as a factual basis, a fine-tuned Large Language Model (LLM) will produce a coherent narrative summary of the findings. This evidence-grounded approach is designed to ensure accuracy and minimize the risk of model hallucination.

Priority: High. This feature delivers the primary actionable output for the clinician and is a key differentiator of the system.

4.3.2 Stimulus/Response Sequences

1. **Stimulus:** A structured JSON file containing quantitative and anatomical tumor data is successfully generated (internal trigger following Feature 4.2).
2. **System Response:**
 - The system automatically sends the JSON data as input to the fine-tuned LLM.
 - The LLM processes the data and generates a multi-paragraph narrative report.
 - The generated text is displayed in an editable panel within the user’s analysis dashboard.
 - The system establishes provenance links between the sentences in the report and the source JSON data.

4.3.3 Functional Requirements

REQ-14: The system shall use the structured JSON file as the sole source of factual information for the report generation process.

REQ-15: The system shall utilize a fine-tuned MedAlpaca-7B language model with Parameter-Efficient Fine-Tuning (LoRA) for generating the narrative.

REQ-16: The generation process shall first populate a deterministic report template with values from the JSON file.

REQ-17: The LLM shall then refine and rephrase the populated template into a fluent, professional, radiology-style narrative.

REQ-18: The final generated report shall be presented to the clinician in an editable text field within the web interface.

REQ-19: The system shall provide a mechanism for the user to trace each key statement in the report back to the specific data point in the JSON file that supports it.

4.4 System Feature 4: Continual Learning and Model Adaptability

4.4.1 Description and Priority

This feature addresses a core limitation of static AI models: performance degradation when encountering new data from different scanners or protocols (domain shift). The system is designed to be a dynamic, evolving tool that can learn from new datasets over time without catastrophically forgetting previously learned knowledge. This is achieved by implementing a continual learning (CL) framework.

Priority: High. This is a key research contribution and a critical feature for the long-term viability and robustness of the system in a real-world clinical environment.

4.4.2 Stimulus/Response Sequences

1. **Stimulus:** A system administrator, with access to the backend, initiates a model update process, providing a new, curated dataset of labeled MRI scans.
2. **System Response:**
 - The system ingests the new dataset and initiates the continual learning training pipeline.
 - The CL algorithm updates the existing segmentation model's weights, incorporating knowledge from the new data while preserving performance on previous tasks.
 - The system evaluates the updated model against a benchmark dataset, logging performance metrics related to both learning and forgetting.
 - A new, versioned model is saved, ready for deployment by the administrator.

4.4.3 Functional Requirements

REQ-20: The system shall provide a backend mechanism for an administrator to introduce new, labeled MRI datasets into the training pipeline.

REQ-21: The system shall implement continual learning strategies to mitigate catastrophic forgetting when training on new data.

REQ-22: The continual learning framework must include an implementation of Elastic Weight Consolidation (EWC), which penalizes changes to parameters critical for previous tasks.

REQ-23: The framework must also include an implementation of replay-based memory, where a small, representative subset of past data is stored and used during retraining.

REQ-24: The system shall support a hybrid approach combining both EWC and replay-based methods.

REQ-25: The system must maintain version control for the segmentation models, allowing an administrator to deploy updated models or roll back to a previous version.

REQ-26: The training pipeline shall log CL-specific performance metrics, such as learning accuracy on new tasks and knowledge retention on old tasks.

4.5 System Feature 5: Clinician-in-the-Loop and Report Finalization

4.5.1 Description and Priority

This feature encompasses the crucial interactive capabilities that allow a clinician to review, validate, and refine the AI-generated outputs. It transforms the system from a simple automated tool into a collaborative diagnostic assistant. The user can edit the narrative report, verify findings through traceability links, provide feedback, and ultimately export a finalized, professional-grade report.

Priority: High. The human-in-the-loop functionality is essential for ensuring clinical accuracy, building user trust, and enabling the system's outputs to be used in a real-world workflow.

4.5.2 Stimulus/Response Sequences

1. **Stimulus:** The user reviews the AI-generated visualizations and draft report on the analysis dashboard.
2. **System Response:** The system allows the user to interact with the report. The user can click on a sentence, and the system highlights the corresponding evidence. The user can type in the text box to make edits.
3. **Stimulus:** The user clicks the "Provide Feedback" button to flag an error in the segmentation.
4. **System Response:** The system presents a simple form to capture the user's feedback and stores it in the database.
5. **Stimulus:** The user, satisfied with the report, clicks the "Export to PDF" button.
6. **System Response:** The system generates a PDF document containing the finalized text and key visuals and initiates a download for the user.

4.5.3 Functional Requirements

REQ-27: The system shall allow the user to directly edit the text of the AI-generated narrative report within the web interface.

REQ-28: The system shall implement an interactive traceability feature where clicking on a key sentence in the report highlights the corresponding evidence in the visualization panels.

REQ-29: The user interface shall include a mechanism for the clinician to provide feedback on the quality of the AI’s output (e.g., flagging an incorrect segmentation).

REQ-30: The system must capture and store all clinician edits and feedback in the database to create a dataset for future model improvement.

REQ-31: The system shall provide a function to export the finalized report.

REQ-32: The exported report must be in the Portable Document Format (PDF).

REQ-33: The final PDF report must include the finalized narrative text, selected 2D slice visualizations with segmentation overlays, and at least one 3D rendering of the segmented tumor.

5 Other Nonfunctional Requirements

This section specifies nonfunctional requirements that are not covered by the system features in Section 4. These include requirements related to performance, safety, security, and overall software quality.

5.1 Performance Requirements

The following requirements define the performance characteristics of the system. They are crucial for ensuring the system is a practical and efficient tool in a clinical support context.

PERF-1: End-to-End Processing Time The total time for the end-to-end workflow—from the completion of a user’s multimodal MRI scan upload to the display of the draft diagnostic report and visualizations—shall be less than 5 minutes for a typical dataset.

- **Rationale:** To be a useful diagnostic aid that helps reduce clinician turnaround time, the system must deliver results in a reasonably short period.

PERF-2: User Interface Responsiveness All user interface interactions, such as loading pages or responding to button clicks, shall complete in under 2 seconds. The UI must never become unresponsive or "freeze" while backend processing is occurring.

- **Rationale:** A responsive UI is critical for a positive user experience and efficient workflow. Asynchronous processing for long-running tasks ensures the clinician can continue to interact with the application.

PERF-3: Inference Latency Optimization The inference latency for both the segmentation model (MoME+) and the language model (MedAlpaca-7B) shall be optimized. This will be achieved through model quantization (e.g., bitsandbytes, GPTQ) and the use of high-performance serving runtimes (e.g., ONNX, vLLM).

- **Rationale:** Efficient model inference is the primary factor in achieving the overall processing time target (PERF-1) and helps to minimize the computational cost of deployment.

PERF-4: Concurrent User Handling The system, in its initial deployed version, must be able to handle at least 3 concurrent analysis requests without significant degradation in the processing time for any individual request.

- **Rationale:** While not expected to handle a heavy load as a prototype, the system architecture must be capable of managing a small number of simultaneous users, as might be expected in a small clinical department or research group.

5.2 Safety Requirements

As a tool intended for use in a clinical support context, the system must adhere to requirements that ensure user and patient safety. The primary safety risk is not direct harm, but the potential for providing incorrect information that could mislead a clinician. The following requirements are designed to mitigate this risk.

SAFE-1: Advisory Role Disclaimer The system’s user interface must, upon startup and within its "About" section, prominently display a disclaimer stating that it is an investigation tool intended for research and informational support only. It must explicitly state that it is **not** a certified medical device and should **not** be used for primary diagnosis.

- **Rationale:** This manages user expectations and reinforces the system’s role as a support tool, ensuring the clinician remains the final diagnostic authority.

SAFE-2: Clinician as Final Authority The system’s design must ensure that the clinician is always in control. All AI-generated outputs, including segmentation masks and reports, must be presented as drafts that require clinician review, editing, and final approval. The system shall not automatically finalize or transmit any report without explicit user action.

- **Rationale:** This human-in-the-loop design is the most critical safety mechanism, preventing the system from making autonomous decisions and ensuring expert oversight.

SAFE-3: Graceful Failure Handling In the event of an internal error during any stage of the analysis pipeline (e.g., model inference fails, data format is invalid), the system must not produce a partial or potentially incorrect result. Instead, it must halt the process for that specific case and present a clear, understandable error message to the user.

- **Rationale:** Prevents the system from presenting flawed or incomplete data as a valid result, which could be dangerously misleading.

SAFE-4: Data Integrity The system must ensure that the input data (MRI scans) are not altered or corrupted during the analysis process. The original uploaded files must be kept intact and separate from the generated outputs.

- **Rationale:** Guarantees that the source data remains in its original state for verification and archival purposes.

SAFE-5: No Control of Medical Devices The software shall not interface with or exert any control over any clinical or life-sustaining hardware or medical devices.

- **Rationale:** Explicitly limits the scope of the system to data analysis, eliminating risks associated with controlling physical hardware.

5.3 Security Requirements

The system will handle sensitive and confidential patient data (MRI scans). Therefore, robust security measures are required to protect this data from unauthorized access, use, or disclosure. The following requirements define the security posture of the system.

SEC-1: User Authentication The system shall implement a mandatory user authentication mechanism. All users must be required to log in with a unique username and password before they can access any part of the system.

- **Rationale:** To ensure that only authorized and identified individuals can access the system and its data.

SEC-2: Data Encryption in Transit All data transmitted between the client's web browser and the backend server must be encrypted using the Hypertext Transfer Protocol Secure (HTTPS) with Transport Layer Security (TLS 1.2 or higher).

- **Rationale:** To protect sensitive data from eavesdropping or man-in-the-middle attacks during transmission over the network.

SEC-3: Data Encryption at Rest All stored MRI scans and patient-related data must be encrypted at rest in the cloud storage (GCP Buckets) and in the database (PostgreSQL).

- **Rationale:** To protect the data from unauthorized access even in the event of a physical or logical breach of the storage infrastructure.

SEC-4: Secure Password Storage User passwords must not be stored in plaintext. The system shall store passwords using a strong, one-way cryptographic hashing algorithm with a salt (e.g., bcrypt).

- **Rationale:** To prevent user account compromise if the database is breached.

SEC-5: Data De-identification Where possible, the system should encourage the use of de-identified data. The user manual will include a section advising users on how to remove personally identifiable information (PII) from NIfTI file headers before uploading.

- **Rationale:** To minimize the privacy risk associated with handling patient data. While the system must be secure, reducing the amount of PII it handles is a primary privacy-enhancing measure.

SEC-6: Session Management The system shall implement secure session management practices, including the use of secure session cookies and timeouts for inactive user sessions.

- **Rationale:** To prevent unauthorized access to the system from an unattended, logged-in terminal.

5.4 Software Quality Attributes

This section defines the key quality attributes that the system must exhibit. These attributes are crucial for the system's success as a reliable, usable, and maintainable clinical support tool.

Adaptability (High Priority): The system's core design must be adaptable to new data. Through its continual learning framework (Feature 4.4), the segmentation model shall be capable of integrating new datasets to improve its performance and handle domain shifts without requiring a complete redesign.

Correctness & Reliability (High Priority): The system must produce accurate and trustworthy results.

- The segmentation model shall achieve high performance on standard validation benchmarks (e.g., Dice, IoU).
- The report generation module's primary quality attribute is reliability; it must not "hallucinate" or generate statements that are not directly supported by the structured JSON evidence.

Usability (High Priority): The user interface must be intuitive and efficient for clinicians with varying levels of technical expertise. A new user should be able to learn the core workflow (upload, view, export) in under 10 minutes. The system is intended to reduce workload, so its design must prioritize clarity and ease of use.

Maintainability (Medium Priority): The system shall be designed for ease of maintenance and future updates. This will be achieved through:

- A modular, microservice-based backend architecture.
- Well-documented code and a version-controlled codebase (Git/GitHub).
- An automated testing and deployment pipeline (CI/CD via GitHub Actions).

Deployability (Medium Priority): The entire application stack (frontend, backend, ML models) shall be containerized using Docker. This ensures that the system can be deployed consistently and reliably across different environments, from a local development machine to the target cloud platform (GCP).

Robustness (High Priority): The system must handle unexpected situations gracefully. This includes invalid file formats, incomplete data uploads, and internal model errors. As defined in SAFE-3, the system must fail safely by providing clear error messages rather than producing incorrect results.

5.5 Business Rules

This section outlines the key operating principles and policies that govern the system’s use. These rules are not functional requirements themselves but they dictate the context in which the functional requirements must be implemented.

BR-1: Clinician’s Final Authority All AI-generated outputs, including segmentation masks and narrative reports, are considered preliminary drafts. A human clinician must review, potentially edit, and explicitly approve the final report. The system is a decision-support tool, and the ultimate diagnostic responsibility lies with the user.

BR-2: Data Isolation Each user account is a separate entity. Data uploaded by one user, and the results generated from it, shall not be visible to or accessible by any other user.

BR-3: Mandatory Multi-modal Input The automated analysis workflow will only be initiated if the user provides the complete set of required multi-modal MRI scans. The system will not attempt to run an analysis on an incomplete or partial dataset.

BR-4: Feedback for System Improvement Any corrections or edits made by a clinician to the AI-generated report shall be considered expert-labeled data. The system is designed to log these corrections, which may be used by system administrators to create new datasets for the continual learning pipeline, thereby improving the model over time.

BR-5: Versioned Models The underlying AI models will be versioned. The version of the segmentation model and LLM used to generate a specific report must be recorded and included in the report’s metadata for traceability and reproducibility.

6 Other Requirements

This section defines any additional requirements and considerations that are not covered in the preceding sections.

Ethical Considerations: The system must be developed with an awareness of potential ethical issues in medical AI. While the training data is sourced from public repositories, it is acknowledged that this data may contain inherent biases. The system’s design incorporates an explainability framework (traceability) and a mandatory clinician-in-the-loop workflow to mitigate the risk of biased or incorrect automated outputs influencing clinical decisions.

Data Privacy and Legal Considerations: Although the system is a prototype and not intended for primary diagnostic use, it is designed with principles of data privacy in mind. All security requirements (Section 5.3), such as data encryption and user authentication, must be strictly followed. The system’s data handling policies should be guided by standards set forth in regulations like the Health Insurance Portability and Accountability Act (HIPAA) concerning patient data confidentiality.

Sustainable Development Goals (SDGs): The project is developed with the explicit goal of contributing to the United Nations Sustainable Development Goals.

- **SDG 3 (Good Health and Well-Being):** The system aims to improve the accuracy, speed, and accessibility of brain tumor diagnosis, ultimately contributing to better health outcomes.
- **SDG 9 (Industry, Innovation, and Infrastructure):** The project fosters innovation in healthcare AI within Pakistan by developing a modern, scalable, and deployable diagnostic tool, strengthening the digital health infrastructure.

Code Reusability and Modularity: The system shall be built using a modular architecture (containerized microservices) to promote the reusability of its components. The goal is for the core segmentation engine, report generation module, and front-end components to be maintainable and potentially adaptable for other medical imaging projects in the future.

Database Requirements

The system requires a relational database to store user data, session metadata, and analysis results. The logical structure of this database is defined by the Entity-Relationship Diagram (ERD) shown in Figure 11.

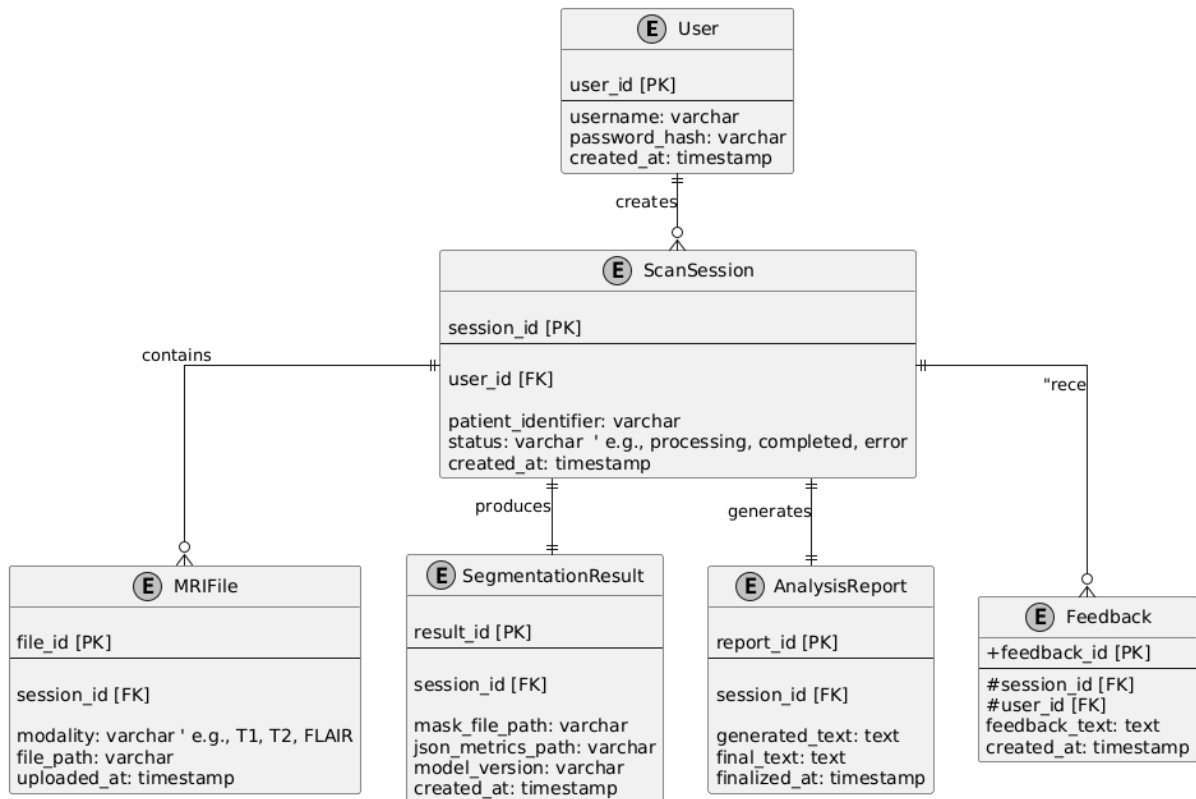


Figure 11: Entity-Relationship Diagram (ERD) illustrating the database schema, including key entities and their relationships.

A Glossary

AI

Artificial Intelligence. A branch of computer science focused on creating systems that can perform tasks that typically require human intelligence. In this project, it refers specifically to deep learning models.

API

Application Programming Interface. A set of rules and protocols that allows different software applications to communicate with each other. This project uses a REST API for client-server communication.

BraTS

Brain Tumor Segmentation Challenge. An annual competition and a primary source of high-quality, publicly available multimodal MRI datasets for brain tumor analysis.

CL

Continual Learning. A machine learning paradigm where a model learns sequentially from new data over time while retaining previously learned knowledge. It is a core feature of this project, designed to mitigate catastrophic forgetting.

CUDA

Compute Unified Device Architecture. An NVIDIA platform and API that allows software like PyTorch to use the processing power of GPUs, which is essential for accelerating deep learning tasks in this project.

ET

Enhancing Tumor. One of the three key subregions of a brain tumor identified during segmentation, typically representing the most active part of the tumor.

EWC

Elastic Weight Consolidation. A specific continual learning algorithm that slows down learning on weights that are deemed important for previous tasks, thus preventing forgetting.

GCP

Google Cloud Platform. A suite of cloud computing services used in this project for data storage (GCP Buckets) and computation (GPU-enabled virtual machines).

GUI

Graphical User Interface. A visual interface that allows users to interact with electronic devices through graphical icons and visual indicators, as opposed to text-based interfaces.

JSON

JavaScript Object Notation. A lightweight, human-readable data-interchange format used for communication between the server and client and as the structured input for the LLM.

LLM

Large Language Model. A type of AI model trained on vast amounts of text data to understand and generate human-like language. This project uses MedAlpaca-7B.

LoRA

Low-Rank Adaptation. A Parameter-Efficient Fine-Tuning (PEFT) method used to fine-tune large models like LLMs by training only a small number of new weights, making the process much more efficient.

MoME+

Mixture-of-Modality Experts. An advanced deep learning architecture for medical image segmentation that uses separate "expert" networks for each imaging modality (e.g., T1, T2) to improve performance.

MONAI

Medical Open Network for AI. An open-source, PyTorch-based framework specifically designed for deep learning in medical imaging.

MRI

Magnetic Resonance Imaging. A medical imaging technique used to create detailed images of the organs and tissues within the body, which is the primary input data for this system.

NIfTI

Neuroimaging Informatics Technology Initiative. A standard file format (.nii or .nii.gz) for storing medical imaging data, particularly in neuroimaging.

SDK

Software Development Kit. A collection of software development tools in one installable package. In this project, an SDK (e.g., the GCP SDK) is used to interface with external services from the application code.

SRS

Software Requirements Specification. The document that describes the functional and non-functional requirements of a software system. This document is an SRS.

TC

Tumor Core. A subregion of a brain tumor that includes the enhancing tumor but excludes the surrounding edema.

WT

Whole Tumor. The largest subregion of a brain tumor, encompassing all abnormal tissue including the tumor core and peritumoral edema.

B To Be Determined List

As of the current version of this document, all requirements have been specified to the best of our knowledge and no items are marked as "To Be Determined" (TBD).